MATH 128a: Homework #3

Instructor: F. Weber

Due by: **Wednesday, September 24, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from textbook Section 1.3

Solve problems 3, 6, 7, 8(13),10(15), 11(16) from Section 1.3 of the textbook.
(6,7: justify your answers.)
If the problem has a different number in edition 9 of the textbook, it is in parentheses.

Answer of Problem 1

- Problem 3:

  a. $|4P_n(1) - \pi| < 10^{-3}$

  Solution: This is an alternating series with terms that decrease in magnitude. It suffices to find n such that

  $$\frac{4}{2(n+1)-1} \leq 10^{-3}.$$

  Hence, we can take $n = 2000$.

  b. $|4P_n(1) - \pi| < 10^{-10}$

  Solution: With the same process as (a), we find that $n = 2 \times 10^{10}$. This is a large number. Depending on the performance of your processor and the language you are using, this could take anywhere from minutes to months. The Madhava–Leibniz formula for π converges very slowly because the behavior is controlled by the inverse linear term rather than the exponential of the Taylor series. There are better ways of computing π, some of which involve modifying this series and others that are largely independent.

- Problem 6:

  a. $\lim_{n\to\infty} \sin\frac{1}{n} = 0$.

  Solution: For any x, we have $\sin x = 0 + R_0(x) = \cos(\xi(x))x$, with $\xi(x)$ between 0 and x. Thus, $|\sin x| \leq |\cos\xi(x)| \cdot |x| \leq |x|$. In particular, $|\sin(1/n)| \leq K|1/n|$ with $K = 1$. So $\sin(1/n) = O(1/n)$.

b. $\lim_{n\to\infty} \sin\frac{1}{n^2} = 0$

Solution: From (a), $|\sin(1/n^2)| \le K|1/n^2|$ with $K = 1$, so $\sin(1/n^2) = O(1/n^2)$.

c. $\lim_{n\to\infty} \left(\sin\frac{1}{n}\right)^2 = 0$.

Solution: Since $y = x^2$ is an increasing function for $x \ge 0$, the result from (a) gives that $|\sin(1/n)^2| = |\sin(1/n)|^2 \le (|1/n|)^2 = K|1/n^2|$ with $K = 1$. So $\sin(1/n)^2 = O(1/n^2)$.

d. $\lim_{n\to\infty} \left[\ln(n+1) - \ln(n)\right] = 0$.

Solution: First, write the difference of logarithms as $\ln\left(1 + \frac{1}{n}\right)$. For $x > 0$, we have $\ln(1+x) = 0 + R_0(x) = \frac{1}{1+\xi(x)}x$, where $\xi(x) \in (0, x)$. Since $\xi(x) > 0$, $1/(1+\xi(x)) < 1$. We conclude that $|\ln(1+x)| \le x$ when $x > 0$. Setting $x = 1/n$ shows that $\left|\ln(1+\frac{1}{n})\right| \le K|1/n|$ with $K = 1$, i.e. $[\ln(n+1) - \ln n] = O(1/n)$.

- Problem 7:

a. $\lim_{h\to 0} \frac{\sin h}{h} = 1$.

Solution: Let $f(x) = \sin x - x$. The first nonzero term in the Taylor expansion is $n = 3$. Back up one and write $f(x) = 0 + 0 + 0 + R_2(x) = \frac{-\cos(\xi(x))}{3!}x^3$. Then for $h \ne 0$, $|f(h)/h| = |\cos(\xi(h))h^2|/6 \le K|h^2|$ with $K = 1/6$. Thus, $\frac{\sin h}{h} - 1 = f(h)/h = O(h^2)$.

b. $\lim_{h\to 0} \frac{1-\cos h}{h} = 0$.

Solution: $1 - \cos x = 0 + 0 + R_1(x) = \frac{\cos(\xi(x))}{2!}x^2$, so $|(1-\cos h)/h| \le K|h|$ with $K = 1/2$. We conclude that $(1 - \cos h)/h = O(h)$.

c. $\lim_{h\to 0} \frac{\sin h - h\cos h}{h} = 0$.

Solution: Note that $\sin x - x\cos x = 0 + 0 + 0 + \frac{2\cos\xi(x) - \xi(x)\sin\xi(x)}{3!}x^3$ with $\xi(x)$ between 0 and x. If $|x| \le 1$ then $|\xi(x)| \le 1$ and $|2\cos\xi(x) - \xi(x)\sin\xi(x)| \le 2|\cos\xi(x)| + |\xi(x)| \cdot |\sin\xi(x)| \le 2 + 1 = 3$. Thus, for $|h| \le 1$, $\left|\frac{\sin h - h\cos h}{h}\right| \le K|h^2|$ with $K = \frac{3}{3!} = 1/2$. Result is $O(h^2)$.

d. $\lim_{h\to 0} \frac{1-e^h}{h} = -1$.

Solution: $1 - e^x + x = 0 + 0 + \frac{-e^{\xi(x)}}{2!}x^2$. Thus, if $|h| \le 1$, then $|\xi(h)| \le 1$ and we have $\left|\frac{1-e^h}{h} + 1\right| = \frac{|e^{\xi(h)}|}{2}|h| \le K|h|$ with $K = \frac{e}{2}$. We conclude that $\frac{1-e^h}{h} = -1 + O(h)$.

- Problem 8:

a. Show that $\alpha_n = \alpha + O(n^{-q})$.

Solution: We are given that there exist $N \ge 1$ and $K < \infty$ such that $|\alpha_n - \alpha| \le Kn^{-p}$ for $n \ge N$. Since $p > q$ and $n \ge 1$, we have $n^{-p} \le n^{-q}$. (Take the log of both sides to confirm.) Thus, for $n \ge N$, $|\alpha_n - \alpha| \le Kn^{-p} \le Kn^{-q}$, as claimed. (Same N and K.)

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Table 1: Table of values for $1/n$, $1/n^2$, etc...

n	5	10	100	1000
$1/n$	0.2	0.1	0.01	0.001
$1/n^2$	0.04	0.01	0.0001	0.000001
$1/n^3$	0.008	0.001	0.000001	0.000000001
$1/n^4$	0.0016	0.0001	0.00000001	0.000000000001

b. Make a table listing $1/n$, $1/n^2$, $1/n^3$, and $1/n^4$ for $n = 5$, 10, 100, and 1000, and discuss the varying rates of convergence of these sequences as n becomes large.

Solution:
The sequences with higher exponents converge faster, consistent with (a).

- Problem 10:

a. $F(x) = c_1L_1 + c_2L_2 + O(x^\gamma)$

Solution: We compute

$$F(x) = c_1F_1(x) + c_2F_2(x) = c_1L_1 + c_2L_2 + c_1O(x^\alpha) + c_2O(x^\beta).$$

It suffices to show that $cO(x^\alpha) = O(x^\alpha)$ and that $O(x^\alpha) = O(x^\beta)$ whenever $\alpha \geq \beta$. The former follows from tucking the constant factor into the bound. The latter is a bit more involved.
Let $f(x) = O(x^\alpha)$. In other words, there exists constants C and $\epsilon > 0$ for which

$$|f(x)| \leq C|x^\alpha|$$

when $|x| < \epsilon$. Let $\eta = \min\{\epsilon, 1\}$. Whenever $|x| < \eta$, we have

$$|x|^{\alpha/\beta} \leq |x|$$

because $\alpha/\beta \geq 1$. We can conclude that

$$|f(x)| \leq C|x^\alpha| = C|x|^\alpha \leq C|x|^\beta = C|x^\beta|.$$

In other words, $f(x) = O(x^\beta)$ as well.

b. $G(x) = L_1 + L_2 + O(x^\gamma)$

Solution: We compute

$$G(x) = F_1(c_1x) + F_2(c_2x) = L_1 + L_2 + O((c_1x)^\alpha) + O((c_2x)^\beta).$$

It suffices to show that $O((cx)^\alpha) = O(x^\alpha)$. This follows from properties of exponents and the absolute value:

$$|(cx)^\alpha| = |c||x^\alpha|.$$

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- Problem 11: **Solution.** We start by observing that

$$x_n = \frac{F_{n+1}}{F_n} = \frac{F_n + F_{n-1}}{F_n} = 1 + \frac{F_{n-1}}{F_n} = 1 + \frac{1}{x_{n-1}}, \qquad (n \geq 1).$$

Assuming $x_n \to x$, the limit laws imply that

$$x = 1 + \frac{1}{x}.$$

Multiplying through by x yields $x^2 - x - 1 = 0$, or $x = \frac{1 \pm \sqrt{5}}{2}$. The negative root is ruled out as all the x_n are positive. So $x = \frac{1+\sqrt{5}}{2}$.

Problem 2: Problems from textbook Section 2.1

Solve problems 1, 4, 12(10), 13 from Section 2.1 of the textbook.
If the problem has a different number in edition 9 of the textbook, it is in parentheses.
Do problem 1 'by hand' (you can use a calculator). For the other problems, you may (if you want to) use the matlab scripts from the course website (or create your own).

Answer of Problem 2

- Problem 1: **Solution.** The function is negative on the left and positive on the right. If it is negative, we have guessed too low, and if it is positive, we have guessed too high. On the interval $[0, 1]$, we can use fixed-point binary fractions to make the binary search easier to write down.

n	a_n	b_n	p_n	$f(p_n)$	Next Direction
1	0.0_2	1.0_2	0.1_2	-0.17048	Right
2	0.10_2	1.00_2	0.11_2	$+0.13434$	Left
3	0.100_2	0.110_2	0.101_2	-0.02039	Right
4	0.1010_2	0.1100_2	0.1011_2	$+0.05632$	Left
5	0.10100_2	0.10110_2	0.10101_2	$+0.01781$	Left
6	0.101000_2	0.101010_2	0.101001_2	-0.00133	Right

We have $p_3 = 0.101_2 = 0.625$.

- Problem 4: **Solution.** In the command window, you can run:

```
>> f = @(x) x.^4 - 2*x.^3 - 4*x.^2 + 4*x + 4;
>> bisection(f, -2, -1, 1e-2)
ans = -1.4141
>> bisection(f, 0, 2, 1e-2)
ans = 1.4141
>> bisection(f, 2, 3, 1e-2)
ans = 2.7266
>> bisection(f, -1, 0, 1e-2)
ans = -0.7266
```

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- Problem 12: **Solution.** Note that there are zeros at -2, -1, 0, 1, and 2. To see where the bisection method converges, you can experiment with a low tolerance. But you should observe what is happening on each interval.

```
>> g = @(x) (x + 2)*(x + 1)^2*x*(x - 1)^3*(x - 2);
>> bisection(g, -1.5, 2.5, 1e-5)
ans = 0
```

Even though there are four zeros on this interval, the bisection method quickly throws away many that are on the wrong side of the first bisection. In this case, we ended up with only $x = 0$ after two iterations.

```
>> bisection(g, -0.5, 2.4, 1e-5)
ans = 2.8610e-06
```

This will converge to 0. You might have thought that the previous interval selected 0 because it was closer to the middle. This is a situation in which the bisection method picks a root near the edge. Because the function is positive at the midpoint, 1 and 2 are discarded immediately after the first iteration.

```
>> bisection(g, -0.5, 3, 1e-5)
ans = 2
```

In this case, the function happens to be negative at the midpoint, and so 0 and 1 are discarded immediately.

```
>> bisection(g, -1.5, 1.75, 1e-5)
ans = -4.7684e-07
```

This will converge to 0. Note that bisection method will never see a root like -1 outside of knife-edge conditions because the sign does not change there.

- Problem 13: **Solution.** We can compute $\sqrt[3]{25}$ as the solution to $x^3 - 25 = 0$ on the interval $[2, 3]$. This is because the cube root is monotonic on the positive reals, in that we have

$$2 = \sqrt[3]{8} \leq \sqrt[3]{25} \leq \sqrt[3]{27} \leq 3.$$

Hence, we can run:

```
>> bisection(@(x) x.^3 - 25, 2, 3, 1e-4)
ans = 2.9240
```

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